

RL on Cartridges

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Abstract

RL on Cartridges: Freezes the base model and applies reinforcement learning directly to a small customer- or task-specific memory module, such as a learned KV-cache cartridge, so it acquires specialized agent skills without modifying the full model.

The high level idea is to apply *online* RLVR on top of the cartridge itself rather than the base policy. This should enable quasi-non-parametric adaption to new tasks, ability to learn from agent traces/environment feedback, and continually adapt to new tasks without forgetting.

We introduce **CARL** (chosen backronym): **Cartridge Adaptation through Reinforcement Learning**.

Introduction and Motivation

We want to learn from agent traces and environment feedback, and continually adapt to new tasks.

Enterprise relevance:

1. Customer-specific adaption

- Instead of $\theta \rightarrow \theta'$, you get $\theta + C_{\text{customer}}$, where every enterprise has its own evolving cartridge.

Base Model (shared)

+

Customer A cartridge

Customer B cartridge

Customer C cartridge

2. Cheaper RL

- Base model = 70B params
- Cartridge = 40M params
- RL scales roughly with the number of trainable parameters and optimizer state. Updating the cartridge only is dramatically cheaper than updating the base model.

3. Fast continual learning

- Changing a cartridge can happen every hour upon the presence of new user data.

4. Isolation

- Updating the model for one customer doesn't mess up the other customer
- A bad RL update only damages one customer's cartridge. It doesn't catastrophically interfere with everyone else.

5. Many parallel variants during inference

- Imagine OpenAI has 100,000 enterprises Each has different preferences. You cannot realistically keep 100,000 full models But you can keep 100,000 cartridges.

Goals and research contributions:

- Enable non-parametric¹ adaption to new tasks
- Learn from environment feedback/agent traces
- Learn OOD tasks from context/information
- Continually adapt to new tasks without forgetting

Related work

Cover:

- PEFT/LoRA work
 - A key question of concern here is whether this idea is truly different than just doing LoRA
 - (Asawa et al. 2025), (Asawa, Dimakis, and Zaharia 2026)
- Context distillation
- Cartridges (Eyuboglu et al. 2025).
 - SFT vs. RL analogy ² You can think of it like SFT + RL, where ordinary cartridges is like the base model that went through SFT and then CARL is like RLVR on top of the that (mid-training vs. post-training).

¹By non-parametric here, we mean that the core base model policy isn't being updated... just the cartridge itself.

²You can think of it like SFT + RL, where ordinary cartridges is like the base model

- Self-distillation

Method

Research question

We’d want to test things like:

- **RQ1 (RL on Cartridges)**: can we improve on a task with just RL’ing the cartridge without modifying the base model itself? That is, can reinforcement learning applied only to an experience cartridge achieve task adaptation comparable to updating the full model or a LoRA adapter?
- **RQ2 (OOD Generalization)**: can we add OOD information/context/hints³ into the cartridge, that doesn’t exist in the base policy, to then reinforce during RLVR? Thereby expanding support/pass@k?
- **RQ3: (Continual Learning)**: can we sequentially train on new tasks via new cartridges without degrading performance or inducing catastrophic forgetting?
 - Could also be good to analyze the weight space geometry/sparsity of the updates (i.e., SFT forgets, RL doesn’t)
- **RQ4 (Self-distillation)**: Cartridge OPSD: Use on-policy self-distillation to continually compile privileged experience into a trainable cartridge while keeping the base model frozen. Compare with RL optimizer.

Experiments

Datasets and benchmarks

- Continual Learning Bench ([Asawa et al. 2026](#))

Results and discussion

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Asawa, Parth, Alexandros G Dimakis, and Matei Zaharia. 2026. “SIEVE: Sample-Efficient Parametric Learning from Natural Language.” *arXiv Preprint arXiv:2604.02339*.

³For more on distilling hints into the policy to induce a learning signal for RL to reinforce, see ([Qu et al. 2026](#)).

- Asawa, Parth, Christopher M Glaze, Gabriel Orlanski, Ramya Ramakrishnan, Benji Xu, Asim Biswal, Vincent Sunn Chen, Frederic Sala, Matei Zaharia, and Joseph E Gonzalez. 2026. “Continual Learning Bench: Evaluating Frontier AI Systems in Real-World Stateful Environments.” *arXiv Preprint arXiv:2606.05661*.
- Asawa, Parth, Alan Zhu, Abigail O’Neill, Matei Zaharia, Alexandros G Dimakis, and Joseph E Gonzalez. 2025. “How to Train Your Advisor: Steering Black-Box Llms with Advisor Models.” *arXiv Preprint arXiv:2510.02453*.
- Eyuboglu, Sabri, Ryan Ehrlich, Simran Arora, Neel Guha, Dylan Zinsley, Emily Liu, Will Tennien, et al. 2025. “Cartridges: Lightweight and General-Purpose Long Context Representations via Self-Study.” *arXiv Preprint arXiv:2506.06266*.
- Qu, Yuxiao, Amrith Setlur, Virginia Smith, Ruslan Salakhutdinov, and Aviral Kumar. 2026. “POPE: Learning to Reason on Hard Problems via Privileged on-Policy Exploration.” *arXiv Preprint arXiv:2601.18779*.